

SHAP, United Kingdom

WMO: 032250

Lat: 54.5022N

Lon: 2.6850W

Elev: 255

StdP: 98.30

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-4.7	-7.7	2.0	-5.7	-5.8	2.4	-3.1	15.5	7.4	14.2	7.3	0.6	220	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	23.0	17.2	21.0	16.1	19.2	15.3	18.0	21.7	16.9	19.9	16.0	18.3	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.2	19.4	15.7	11.5	18.3	14.8	10.9	17.2	51.5	21.7	48.5	20.0	45.5	18.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.2	9.8	DB	-9.6	25.6	2.6	1.8	-11.5	26.9	-13.0	27.9	-14.5	29.0	-16.4	30.3	(4)
(5)			WB	-9.7	19.9	2.5	1.1	-11.4	20.6	-12.9	21.3	-14.3	21.9	-16.1	22.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	3.1	3.3	4.4	6.5	9.1	11.9	14.0	13.6	11.3	8.7	5.6	3.5		
	DBStd	4.91	3.27	3.18	3.01	2.81	3.04	2.49	2.20	2.29	2.41	3.07	3.32	3.88		
	HDD10.0	1162	214	189	174	110	52	8	1	1	13	61	136	203		
	HDD18.3	3798	473	422	433	356	285	192	136	147	212	299	383	461		
	CDD10.0	408	0	0	1	4	26	66	124	114	51	20	2	1		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	42	0	0	0	0	1	6	19	15	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	4.0	4.8	4.8	4.5	3.9	3.8	3.5	3.2	3.3	3.5	4.1	4.1	4.6		
(15) Precipitation	PrecAvg	1421	171	129	98	81	85	81	95	104	104	139	155	180		
	PrecMax	1855	268	340	215	154	151	189	227	263	195	264	413	512		
	PrecMin	1025	29	23	33	30	20	30	51	19	21	46	64	43		
	PrecStd	215	71	85	39	28	40	40	38	50	39	59	74	100		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	11.3	14.3	19.1	22.3	24.0	25.5	25.3	20.7	17.0	13.6	11.6		
		MCWB	9.3	9.0	9.7	12.1	16.4	17.4	18.5	18.1	16.7	13.7	12.2	10.6		
	2%	DB	9.6	9.7	11.7	15.2	19.6	21.1	23.3	22.0	18.3	15.2	12.3	10.7		
		MCWB	8.5	8.2	8.9	10.8	14.5	16.0	17.4	17.3	15.1	13.6	11.2	9.7		
	5%	DB	8.5	8.7	10.1	13.0	17.0	19.0	21.2	19.6	16.8	14.2	11.3	9.7		
		MCWB	7.5	7.4	7.9	9.7	12.9	14.8	16.4	15.9	14.0	12.8	10.3	8.8		
	10%	DB	7.6	7.8	8.9	11.4	14.7	17.0	19.1	18.0	15.6	13.3	10.4	8.7		
		MCWB	6.5	6.6	7.2	8.7	11.4	13.5	15.5	15.1	13.3	12.0	9.3	7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.6	10.9	13.1	17.1	18.7	19.4	19.0	17.6	15.1	12.6	11.0		
		MCDB	10.2	10.6	13.3	17.8	21.5	22.5	23.9	23.2	20.0	15.9	13.2	11.3		
	2%	WB	8.7	8.6	9.3	11.4	14.9	16.6	18.1	17.6	15.9	14.1	11.5	10.0		
		MCDB	9.4	9.5	11.1	14.2	18.8	20.2	22.1	20.9	17.4	14.8	12.1	10.6		
	5%	WB	7.7	7.7	8.4	10.2	13.3	15.1	16.9	16.7	14.7	13.1	10.5	9.0		
		MCDB	8.5	8.5	9.7	12.5	16.2	18.2	20.2	19.0	16.2	14.0	11.2	9.7		
	10%	WB	6.7	6.8	7.5	9.2	12.0	14.0	15.9	15.7	13.7	12.1	9.4	7.9		
		MCDB	7.5	7.7	8.8	11.1	14.4	16.5	18.5	17.4	15.2	13.1	10.3	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	6.7	8.2	8.8	8.3	8.4	8.0	7.7	6.5	5.6	5.4		
		MCDBR	5.0	6.1	8.7	11.3	13.0	12.3	13.1	11.3	9.5	6.8	5.8	5.4		
	5% WB	MCWBR	5.2	5.2	6.4	7.7	8.5	8.0	7.9	7.2	6.5	5.6	5.5	5.4		
		MCWBR	5.1	5.3	7.4	10.0	11.4	11.0	10.9	9.3	7.6	5.8	5.5	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.326	0.353	0.373	0.377	0.375	0.376	0.364	0.349	0.334	0.310	0.298		
	taud		2.285	2.369	2.316	2.286	2.319	2.348	2.348	2.394	2.442	2.441	2.360	2.260		
	Ebn at Noon		622	761	818	851	868	872	863	853	819	743	628	552		
	Edh at Noon		58	75	98	115	118	116	114	103	86	69	54	48		
(44) All-Sky Solar Radiation	RadAvg		0.55	1.23	2.35	3.74	4.86	4.91	4.75	3.82	2.70	1.44	0.70	0.42		
	RadStd		0.07	0.14	0.21	0.37	0.37	0.48	0.49	0.28	0.24	0.18	0.07	0.04		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page